

PROJECT BUTTON HR–V0 DXL PROTECTION CARRIER

Single–channel R155 measurement carrier; assemble three controlled variants; no robot–baseline change.

01 Single–channel DXL branch protection core

File: 01\_protection\_core.kicad\_sch

02 UVLO and OVLO threshold candidates

File: 02\_threshold\_dividers.kicad\_sch

03 Bypass, slew and transient candidates

File: 03\_bypass\_and\_transients.kicad\_sch

04 Measurement and diagnostic points

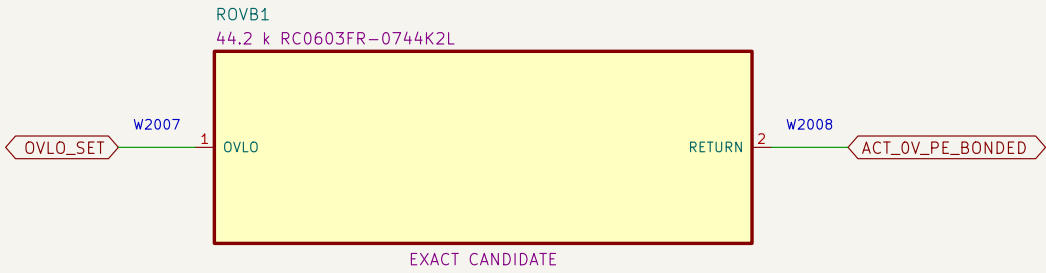
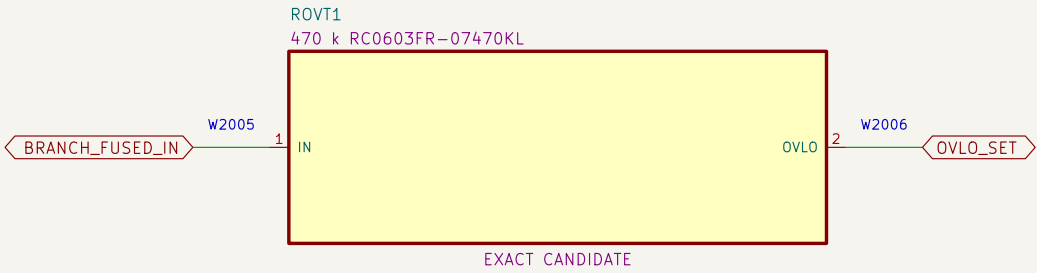
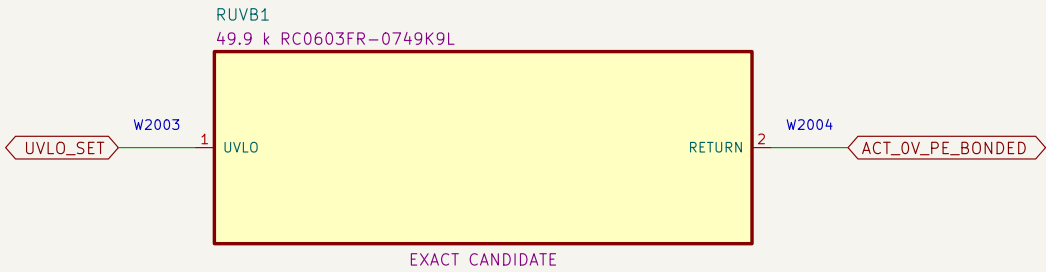
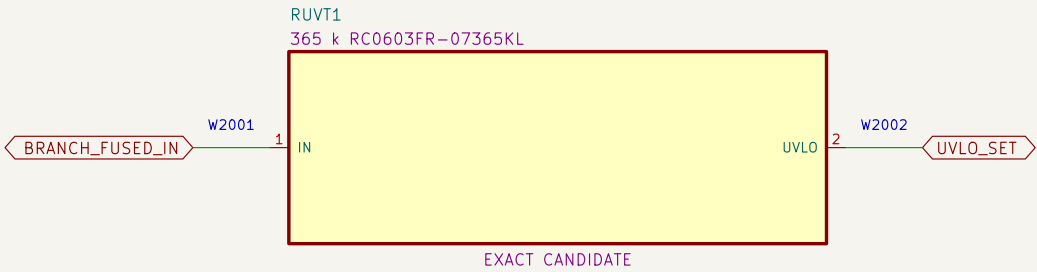
File: 04\_measurement\_points.kicad\_sch

|                                                            |                  |                          |
|------------------------------------------------------------|------------------|--------------------------|
| V2.1 PRESERVED; V3 IS A CONNECTED CANDIDATE                |                  |                          |
| PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION |                  |                          |
| Project Button                                             |                  |                          |
| Sheet: /                                                   |                  |                          |
| File: hr–v0–dxl–protection–carrier.kicad_sch               |                  |                          |
| Title: PROJECT BUTTON HR–V0 DXL PROTECTION CARRIER index   |                  |                          |
| Size: A3                                                   | Date: 2026–08–09 | Rev: DXL–PROT–CARRIER–PO |
| KiCad E.D.A. 10.0.5                                        | Id: 1/5          |                          |



02 UVLO and OVLO threshold candidates

Exact resistor identities; divider tolerance and physical thresholds remain open.



NOTE 1: RUVT1/RUVB1 create the nominal UVLO candidate; ROVT1/ROVB1 use TI's 14 V application-example values.

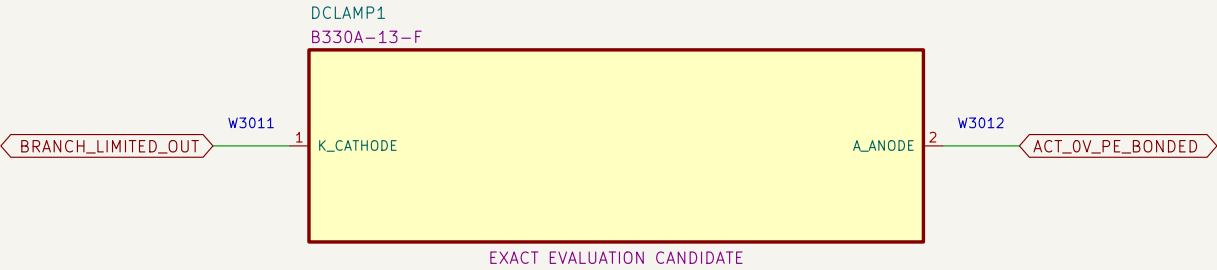
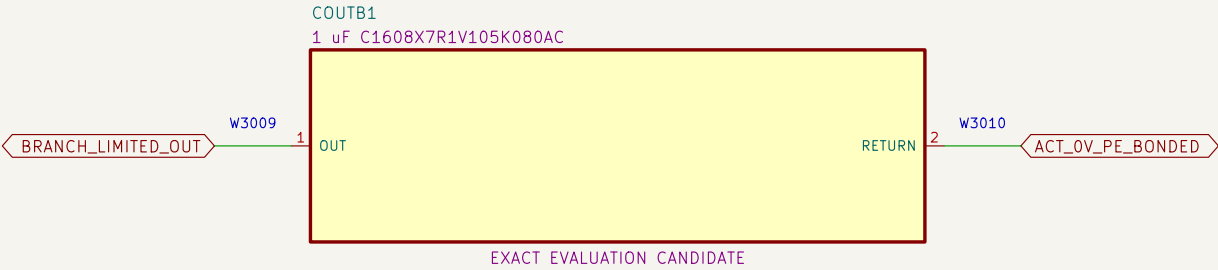
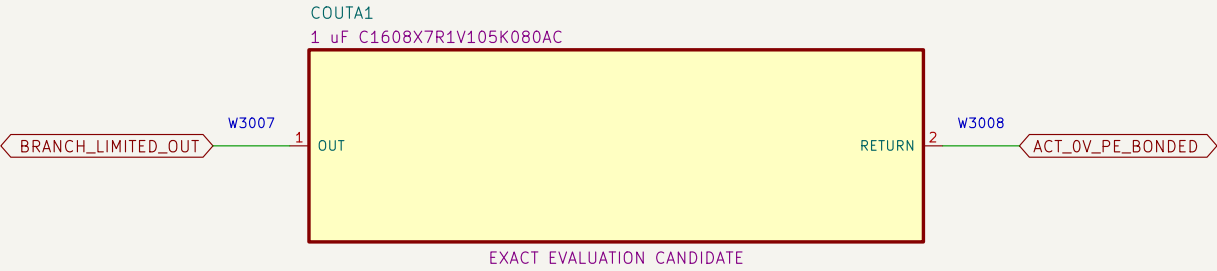
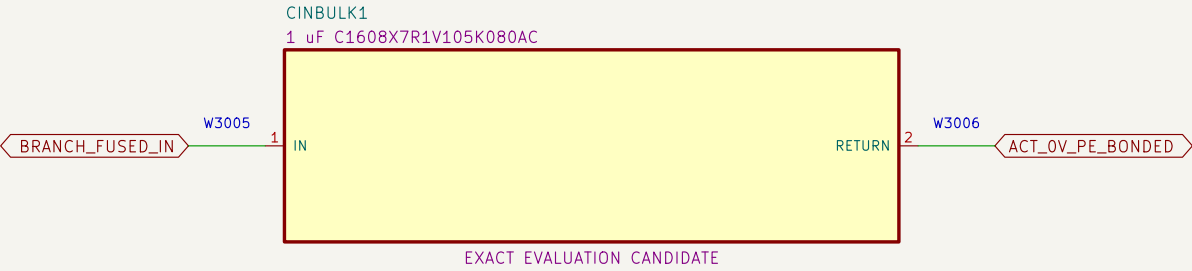
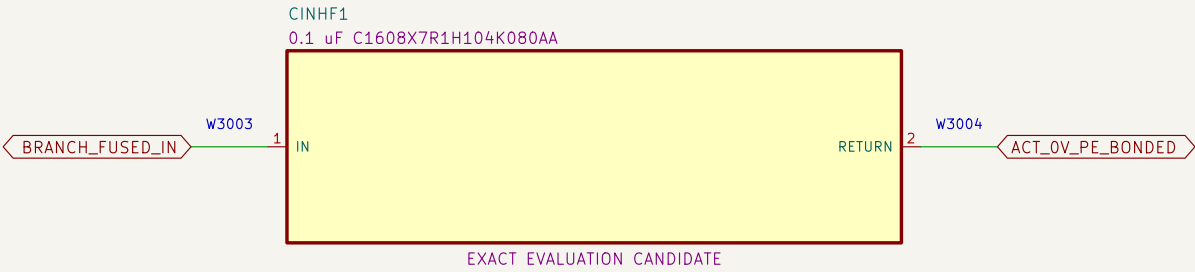
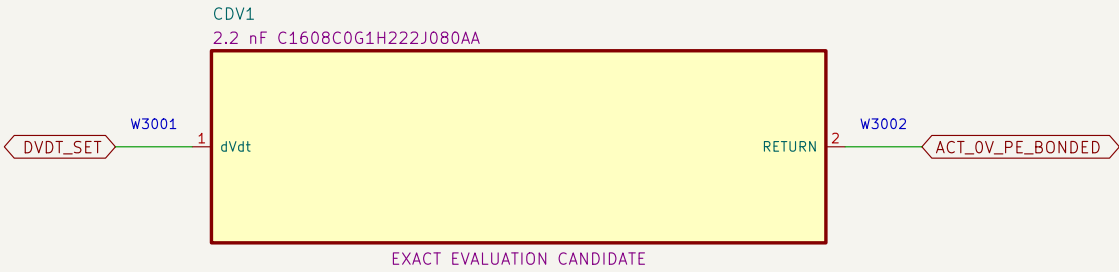
NOTE 2: Received resistor value, tolerance, temperature drift and measured thresholds remain test gates.

NOTE 3: PRELIMINARY – NO SUPPLIER UPLOAD, FABRICATION, CONNECTION, OR ENERGIZATION

|                                                                                         |                  |                            |
|-----------------------------------------------------------------------------------------|------------------|----------------------------|
| CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED                                  |                  |                            |
| PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION                              |                  |                            |
| Project Button                                                                          |                  |                            |
| Sheet: /02 UVLO and OVLO threshold candidates/<br>File: 02_threshold_dividers.kicad_sch |                  |                            |
| Title: PB HR-VO DXL-PROT-CARRIER-P0.1 - 02                                              |                  |                            |
| Size: A3                                                                                | Date: 2026-08-09 | Rev: DXL-PROT-CARRIER-P0.1 |
| KiCad E.D.A. 10.0.5                                                                     | Id: 3/5          |                            |

03 Bypass, slew and transient candidates

Exact passive identities; physical transient, effective–capacitance and thermal proof remain open.



NOTE 1: CINHF1 and CINBULK1 implement TI input guidance; COUTA1/COUTB1 provide nominal output–capacitance margin only.

NOTE 2: DCLAMP1 is the local output Schottky. The external regenerative shunt candidate is not installed on this carrier.

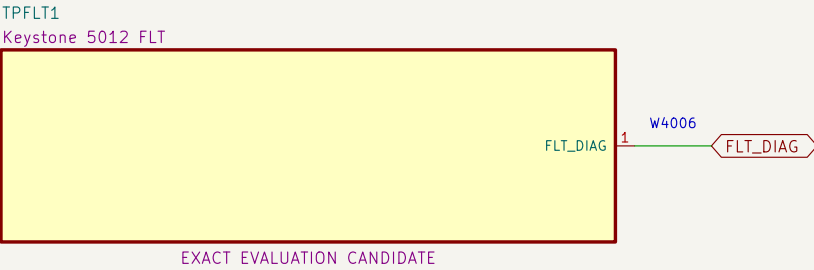
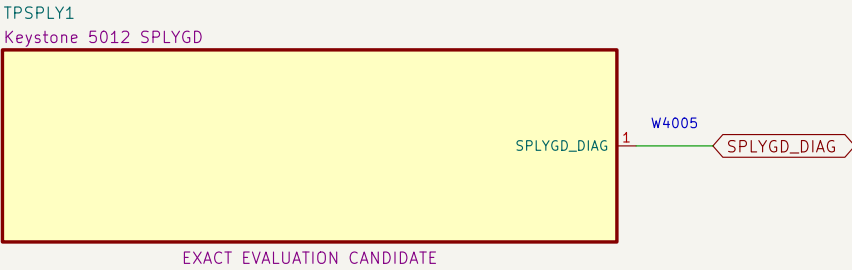
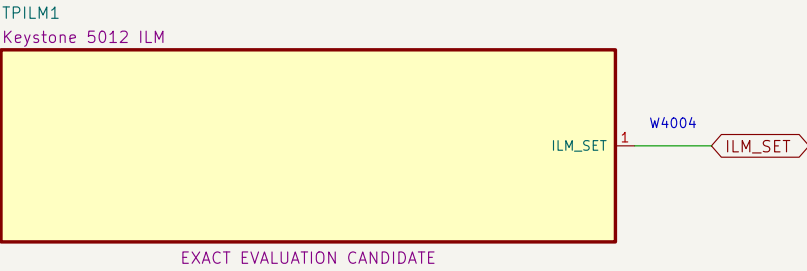
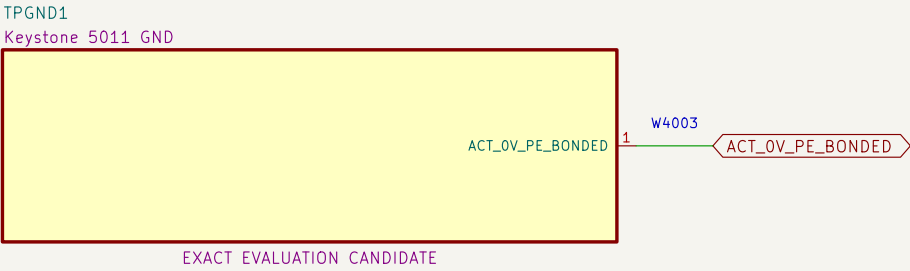
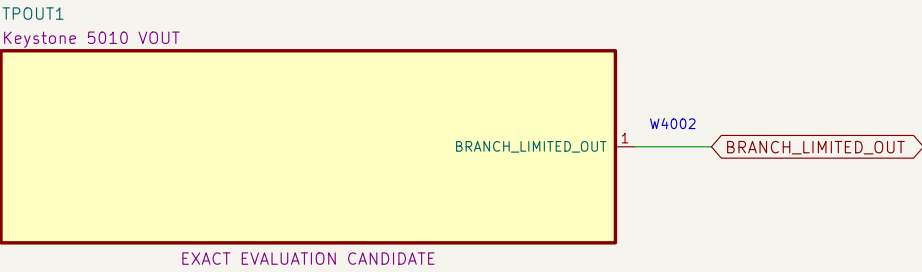
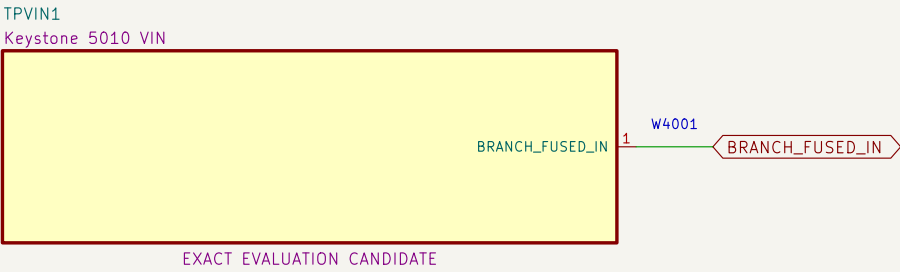
NOTE 3: TVS need and identity remain SELECTION REQUIRED pending actual interconnect–

NOTE 4: PRELIMINARY – NO SUPPLIER UPLOAD, FABRICATION, CONNECTION, OR ENERGIZATION

|                                                                                               |                  |                            |
|-----------------------------------------------------------------------------------------------|------------------|----------------------------|
| CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED                                        |                  |                            |
| PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION                                    |                  |                            |
| Project Button                                                                                |                  |                            |
| Sheet: /03 Bypass, slew and transient candidates/<br>File: 03_bypass_and_transients.kicad_sch |                  |                            |
| Title: PB HR–VO DXL–PROT–CARRIER–P0.1 – 03                                                    |                  |                            |
| Size: A3                                                                                      | Date: 2026–08–09 | Rev: DXL–PROT–CARRIER–P0.1 |
| KiCad E.D.A. 10.0.5                                                                           | Id: 4/5          |                            |

04 Measurement and diagnostic points

All measurements require a released method; no diagnostic has safety or motion authority.



NOTE 1: ILM total parasitic and probe capacitance must remain below TI's 50 pF limit.

NOTE 2: SPLYGD and FLT are unpulled open-drain observations only; no safety, reset or motion credit is assigned.

NOTE 3: Test-plan and data records are blank. No physical test has been executed.

NOTE 4: PRELIMINARY – NO SUPPLIER UPLOAD, FABRICATION, CONNECTION, OR ENERGIZATION

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED  
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION  
**Project Button**  
Sheet: /04 Measurement and diagnostic points/  
File: 04\_measurement\_points.kicad\_sch

**Title: PB HR-V0 DXL-PROT-CARRIER-P0.1 – 04**

|                     |                  |                            |
|---------------------|------------------|----------------------------|
| Size: A3            | Date: 2026-08-09 | Rev: DXL-PROT-CARRIER-P0.1 |
| KiCad E.D.A. 10.0.5 | Id: 5/5          |                            |